

Lunar Lander Analysis Report

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- Effect of epsilon decay on learning speed versus final performance

In Var 1 (higher lr), we can see the reward climb up very fast. The epsilon decay is the same, but higher learning rate helps the agent to get the good experience quicker in a limited amount of time.

But in Var 3, the reward is very flat for a long time. This means if we decay epsilon but the learning signal is bad, the agent just doesn't improve or learn anything new since it no longer takes as much random actions and keeps doing the same wrong things because it stops exploring too early.

- Impact of updating the target network too often or too rarely

If we update too rarely, the agent will have a staggered learning rate because of how the loss is calculated. Like in Var 3, the slower target update is not good. If you look at the average Qvalues, it is totally flat or even negative for 1000 steps. This is because the target network is too old, so the policy network is chasing a outdated goal, trying to minimize an old value that no longer represents the current environment.

But if we update too often, the training might become very unstable and the loss will fluctuate wildly, preventing the values from converging properly.

- Which hyperparameter mattered most

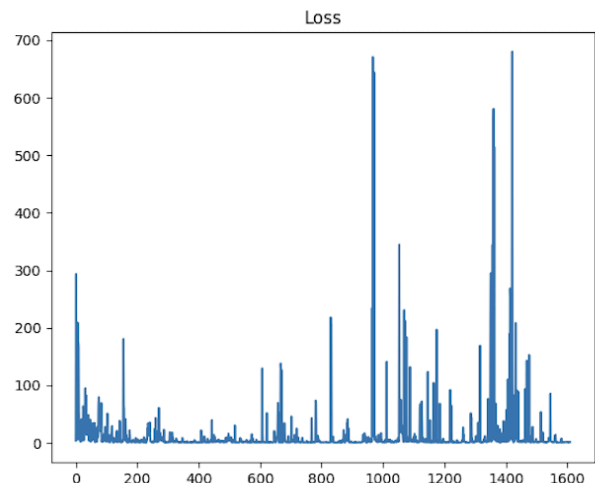
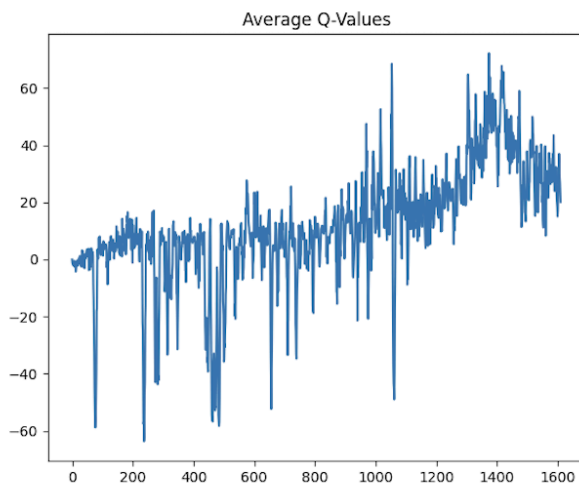
I think learning rate is the most important. If the step size is too small, the weights won't be able to keep up. If the step size is too big, the values won't be able to settle down and converge. Only Var 1 can reach the +200 reward area in the evaluation histogram in my experiments.

I think memory depth also deserves an honorable mention. Var 2 had a smaller memory buffer, and we can see the reward started to drop at the

end because the agent forgot the old successful landing, instead focusing too much on recent experiences and losing the ability to generalize. However, without a good learning rate, the buffer size won't matter as much because the agent never learned how to land in the first place.

- Observed failure modes during training

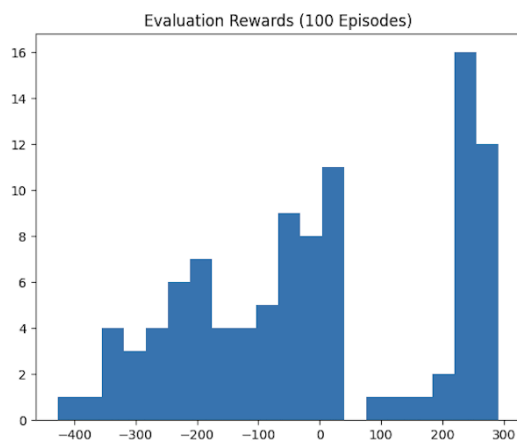
- Random Qvalue Dips (Left Graph): In my opinion, this means the agent is learning to cut its losses and crash early to avoid the penalties for flying for an extended amount of time and using up fuel, acting pessimistic, which is not the goal we wanted the agent to achieve.



- Catastrophic Forgetting (Right Graph): In the graph for Var 2, the loss spiked greatly towards the end. I think that indicates that the agent forgot the broader policy that it learned and focused too greatly on the newer data.

- How the learned policy compares with intuition about optimal landing

- Intuitively, to land a spaceship, you should go down slowly and balance correctly using the side engines, hovering first before descending carefully.
- Learned Policy (Var1): The agent passed the 250 threshold multiple times, successfully learning to land.



- Learned Policy (Var2 Right Graph, Var3 Left Graph): The agent scored -50 and -100 multiple times, so it did not learn to land. It only crashes more slowly to do better than the baseline, surviving longer.

